

Date	16 July 2026
Project Name	Smart Lender AI: An Intelligent Machine Learning System for Loan Approval Prediction
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Maximum Marks	3 Marks

REQUIREMENT ANALYSIS: SOLUTION REQUIREMENTS

Building on the empathy mapping and customer journey stages, the specific technical and operational boundaries of the **Smart Lender AI** platform have been structured into defined functional and non-functional requirements. These requirements ensure that the classification system is both high-performing and user-centric.

1. Functional Requirements (FR)

Functional requirements describe the core capabilities, operations, and services the application must perform to successfully process loan evaluation requests.

REQ ID	REQUIREMENT NAME	DESCRIPTION	PRIORITY
FR-01	Applicant Data Input	The system must provide an interactive, structured interface to collect applicant demographics, income parameters, credit scoring, co-applicant variables, and loan term demands[cite: 10, 11].	High
FR-02	Input Validation & Boundary Safeguards	The system must validate input types dynamically (e.g., non-negative numerical properties and strict length parameters) prior to preprocessing to block invalid requests.	High
FR-03	Dynamic Preprocessing Pipeline	The backend must handle missing value imputations and apply consistent min-max or standard scaling based on historical training bounds.	High
FR-04	Inference Processing	The application must run preprocessed vectors through serialized model pickles (XGBoost/ Random Forest) to calculate loan approval categories and decision confidence levels[cite: 10].	High
FR-05	Visual Analytical Dashboard	The system must display an interactive, visually clear decision dashboard showing key weighted	Medium

REQ ID	REQUIREMENT NAME	DESCRIPTION	PRIORITY
		factors contributing directly to the applicant's prediction outcome[cite: 10, 11].	

2. Non-Functional Requirements (NFR)

Non-functional requirements outline quality constraints, architectural expectations, performance parameters, and usability standards.

REQ ID	REQUIREMENT NAME	DESCRIPTION	PRIORITY
NFR-01	Performance (Latency)	The inference API routing pipeline must return structured predictions and dashboard visualization models in under 1 second of submission[cite: 10].	High
NFR-02	Availability & Scalability	The backend flask architecture must run within isolated environments to support consistent availability during application traffic spikes.	Medium
NFR-03	Portability & Modularity	The system must separate preprocessing rules, pipeline definitions, and classification weight pickles to allow independent system maintenance.	High
NFR-04	Data Security & Privacy	The API must sanitize and process numerical parameters within active memory without storing sensitive demographic details on unencrypted data logs.	High

Architectural Alignment:

These structured requirements translate directly from the **Customer Journey Map**[cite: 10, 11]. For example, the user's emotional anxiety during systemic wait-times is addressed by ****NFR-01** (Under 1-second response latency)**[cite: 10], while "black-box" concerns are mitigated by ****FR-05** (Visual Analytical Dashboard)**[cite: 10, 11].